

Jamie Moritz Holdcroft

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EDUCATION

University of New South Wales (UNSW)

Expected Dec 2027

Bachelor of Science (Mathematics Major, Physics Minor) | Bachelor of Computer Science

Sydney, Australia

- Weighted Average Mark (WAM): **82.87** Distinction, GPA: **6.33/7.0** (UAC methodology)
- **Selected Advanced Coursework (Postgraduate-level):** Extended Algorithm Design & Analysis (HD), Computer Vision (HD), Machine Learning & Data Mining (DN)

Technical University of Munich (TUM)

Apr–Sep 2026

Visiting Student (Exchange)

Munich, Germany

- One-semester exchange in the Department of Mathematics
- **Selected Coursework:** Mathematical Modelling and Simulation, Measure-theoretic Probability Theory and Statistics

Reddam House Sydney, Australia

2018–2023

Higher School Certificate (ATAR: 95.9)

Sydney, Australia

- **Leadership and Awards:** Australian Defence Force (STEM) Future Innovators Award (2023), Liebman Sustainability Award (2023), Head of Environmental Portfolio, Elected to School Executive Representative Council

PUBLICATIONS & RESEARCH OUTPUT

Holdcroft, J., Abdallah, A., & Jatowt, A. (2026). *The Magnitude Mirage: Rethinking Confidence for Reasoning-Intensive Retrieval*. Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026, Main Conference). Accepted

Holdcroft, J., Abdallah, A., & Jatowt, A. (2026). *Difficulty-Gated Fusion of Reasoning Views for Temporal Retrieval*. Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM 2026, Main Conference). Accepted

Abdallah, A., **Holdcroft, J.**, Ali, M., & Jatowt, A. (2026). *Are LLM-Based Retrievers Worth Their Cost? An Empirical Study of Efficiency, Robustness, and Reasoning Overhead*. Proceedings of the 49th International ACM SIGIR Conference (SIGIR 2026, Main Conference). Published

Abdallah, A., **Holdcroft, J.**, Jatowt, A., *et al.* (2026). *Rankify: A Comprehensive Python Toolkit for Retrieval, Re-Ranking, and Retrieval-Augmented Generation*. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026 System Demonstrations Track). Published

Holdcroft, J., Howison, S., & Dewynne, J. (2026). *Self-Referential Forward-Start Options*. **In Preparation**

AWARDS & SCHOLARSHIPS

CSIRO Undergraduate Vacation Studentship

Nov 2026–Feb 2027

12-week full-time research placement, CSIRO Technology / Data61. Selected for the Verifiable Technology Landscape Engine project (Neuro-symbolic verification of technology claims from global corporate filings). Supervised by Dr Haohui Chen.

ARC Undergraduate Research Scholarship

Sep–Nov 2026

Australian Research Council (ARC) Centre of Excellence for the Weather of the 21st Century. 6-week full-time research placement hosted at the UNSW Climate Change Research Centre, evaluating high-resolution CABLE and JULES land surface model simulations across Australia. Supervised by Dr Anna Ukkola (ARC DECRA Fellow) and Dr Mathew Lipson.

EMNLP 2026 Diversity & Inclusion Subsidy

2026

Travel award to present a first-authored main-conference paper in Budapest at EMNLP 2026.

DAAD STIBET Scholarship

2026

German Academic Exchange Service, awarded for exchange at the Technical University of Munich.

EXPERIENCE

Undergraduate Research Scholar

Sep–Nov 2026

ARC Centre of Excellence for the Weather of the 21st Century, UNSW Climate Change Research Centre

- Evaluating high-resolution simulations from CABLE and JULES, the two land surface models used in Australian climate modelling, against Eddy Covariance flux tower observations of energy, carbon and water exchange

- Working with large climate datasets (NetCDF) in Python to benchmark model performance across Australian sites and climate conditions
- Supervised by Dr Anna Ukkola (ARC DECRA Fellow) and Dr Mathew Lipson.

Research Intern (Information Retrieval & Machine Learning)

Jan–Apr 2026

University of Innsbruck (UIBK), Digital Science Center, Austria

- First-authored two papers accepted to EMNLP 2026 and CIKM 2026; co-authored two further papers at ACL 2026 and SIGIR 2026 with PhD researchers and faculty.
- Contributed to Rankify, an open-source retrieval and RAG toolkit (680+ GitHub stars, published at ACL 2026 System Demonstrations).
- Evaluated systems with new and standard IR metrics, analysing trade-offs in quality, robustness, and efficiency.

Research Project: Self-referential Forward-Start Options

Dec 2025–Present

Research Collaboration with University of Oxford Faculty

- First author of a theoretical paper on pricing self-referential forward-start options with Oxford researchers.
- Led the analysis of the self-referential forward-start digital put, under a manipulation-arbitrage framework.
- Proved, using the unimodality of the option's price-to-strike ratio, that every non-trivial self-consistent strike can be arbitrated, leaving only the worthless contract, which is itself arbitrage-free only below a critical multiplier.

Research Project: Quantum Approximate Optimisation Algorithm

Sep–Nov 2025

University of New South Wales (UNSW), Sydney

- Studied parameter concentration phenomena in the QAOA for the Max-Cut problem across graph classes.
- Built a scalable grid-search pipeline for systematic exploration and benchmarking of QAOA parameter landscapes.
- Group supervised by Dr. Gerald Huang.

Casual Academic

Jan 2025–Dec 2025

UNSW Computer Science and Engineering

- Recommended by lecturer and head academic staff to lead weekly undergrad laboratory classes in C programming.
- Collaborated effectively with academic staff to achieve 100% satisfaction rate in the MyExperience survey.

Independent School Canteen Operator

Reddam House Sydney, Australia

Jan 2022–Aug 2023

- Co-founded and independently ran the school canteen while an active student catering to 350+ staff and students.
- Generated national media coverage, featuring on 9 News Australia, 2GB Radio, Radio NZ, the Sydney Morning Herald and more.

SELECTED PROJECTS

- **Time-Series Forecasting (ML):** Developed ensemble models (RF, Ridge, GB) with feature engineering (EW-MAs, lags, FFT, PCA) for multi-horizon prediction on environmental climate data.
- **Dead-Tree Segmentation from UAV RGB-IR Imagery (CV):** Developed and compared rule-based, Random Forest on superpixels, and U-Net CNN approaches; achieved 67% IoU on aerial imagery.
- **Navigation System (C):** Implemented Dijkstra-based routing on road-maps with time-dependent constraints and dynamic road closure updates.
- **Huffman Compression (C):** Built an end-to-end lossless compression system using optimal prefix-free encoding.

TECHNICAL SKILLS

Languages & Tools: C, Python, Bash, Linux, Git, Jupyter, L^AT_EX, Maple

ML / AI Frameworks: PyTorch, HuggingFace, scikit-learn, OpenCV, NumPy

INTERESTS

Poetry, philosophy, climate and marine science, surfing, hiking, jazz